Post-Quantum Cryptography

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***Abstract*— As the development of computers follows Moore's Law, which is we can expect a higher speed and capability improvement every couple of years. The industry has estimated the materials may face their limit in the next decades. But, with the huge effort of research on the quantum computer thesis, new amazing technology is starting to show the next generation development of quantum computer becomes possible. This paper will focus on the post-quantum cryptography, which is what kind of impact will quantum computer has on cryptography.**

Keywords— quantum computing, post-quantum cryptography, public-key cryptography, quantum-resistant

I. Introduction

This paper starts with what is a quantum computer and the difference between the quantum computer and the traditional computer. Before heavily diving into the thesis and cases, the introduction of quantum computers is necessary to understand the background of this paper. Then, it goes over the encryption method these days and the main cryptography method and talks about why could the quantum computer could put these algorithms at risk. After the risk analysis, it will shortly compare the possible quantum-resistant ciphers. In the end, our group will discuss the current NIST's effort on developing a new standardization of Quantum Resist Cipher for the industry.

II. Quantum Computers

The information-technology is based on bits, and a bit is either 0 or 1 in the binary system. And quantum uses the physical system, which is different from our classical computer, the basic unit of memory in a quantum computer is a quantum bit or qubit rather than a bit. The qubit can have multiple states, such as the spin direction of an electron or the orientation direction of a photon, such as it can represent as 1 and 0 simultaneously. With the help of physical systems, we can represent a lot of possibilities synchronously. It means that the traditional computers may go through possible solutions one at a time in cycles to find an answer, but a quantum computer can have many possible solutions to run at once, so it can find an answer quickly. But, also given the property of the physical system, the quantum computer is not stable, the high sensitivity such as heat and electromagnetic fields can cause more collusion between quantum can lead to a crash. [5]

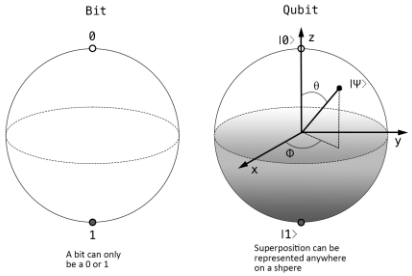


Fig. 1. Difference between Bit and Qubit.

The last piece of the purpose of building the quantum computer is the Riemann hypothesis, if the hypothesis can be proved and have a set of procedures with formula, then finding a target prime number is possible. And given the consideration of prime numbers is applied to cryptography, because the law of prime numbers is not been discovered. But, after the quantum computer is fully developed, it might crack the function of prime, and leave huge security problems in the encryption algorithm whose method is based on mathematics.

III. Risk to Cryptography

Cryptography is a black box with algorithms based on mathematical theory. Most of our cryptography relays on the public key and private key to do encryption or decryption. In short, given the property of qubit, we can try multiple possible solutions once, it will magnificently short the time to do the brute-force attack. Based on the encryption algorithm properties, if the Riemann Hypothesis has proved and find the prime number is easy, then the quantum computer is able to crack the product of a large prime and get the factors. [1]

And our security level is based on the length of the key and the encryption algorithm. It doesn't matter what encryption method we use; quantum computers have a better hash rate to work on some particular problems, such as brute-force attacks. And right now, we could not find a possible way to protect this. Since our password is static, so it find target eventually. In the past, time of that is usually longer than the lifetime of that password, but the quantum is able to shorten it into lifetime, which means after breaking the password, we can still use it to perform some functions or even cause some damage. As for the encryption algorithm, for any symmetric encryption, brute force has a higher possibility to get the key, and for any asymmetric encryption, the Riemann Hypothesis will give the prime number in a possible life-cycle time.

RSA encryption is the most common form of public-key cryptography, it is at risk. In RSA, we use two large prime to get the public key, and the private key is the factor of the large integer. The security level is heavily dependent on the length of the key. The key to getting the product of two large prime numbers is easy but getting the factor of a large integer is hard. Although there have been no known attacks on RSA yet. In 1994, researchers theorized how one might execute an attack, but there was no computer with sufficient computing power to do it. But now quantum computing may realize that. It has the ability to perform the brute-force attack. And given the consideration of the qubit ability and the theoretical Shor's algorithm, it may have been close enough to break

IV. QUANTUM RESISTANT CIPHERS

Public key cryptography has a crucial position in online communication infrastructure. Digital signatures and key establishment are the two most important uses of public-key encryption. As mentioned in Section III, the advent of scalable quantum computing would put these asymmetric key cryptosystems at huge risk. On the contrary, the impact on symmetric encryption will not be such radical (see Table 1). It has been shown that exponential speed-up for searching algorithms is not possible, so symmetric algorithms and hash functions can secure the safety even quantum era comes.

TABLE 1. IMPACT OF QUANTUM COMPUTING ON COMMON CRYPTOGRAPHIC ALGORITHMS [4]

Cryptographic Algorithm	Type	Purpose	Impact from large-scale quantum computer
AES	Symmetric key	Encryption	Larger key sizes needed
SHA-2, SHA-3	----------	Hash functions	Larger output needed
RSA	Public key	Signatures, key establishment	No longer secure
ECDSA, ECDH (Elliptic Curve Cryptography)	Public key	Signatures, key exchange	No longer secure
DSA (Finite Field Cryptography)	Public key	Signatures, key exchange	No longer secure

Therefore, searching for quantum-resistant ciphers for asymmetric algorithms becomes urgent. According to [2], a group of families of post-quantum ciphers has been proposed, including those based on multivariate polynomials, codes, and lattices, as well as a few others, as illustrated in [Fig. 2]. However, none of the proposals has proven to guarantee security against all quantum threats and attacks. A lot more analysis and research need to be done before quantum computers come to reality.

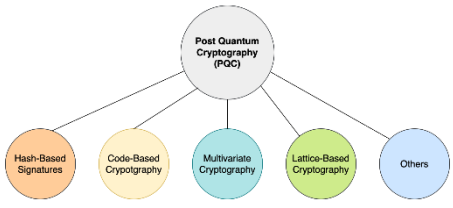


Fig. 2. Families of Quantum Resistant Ciphers

V. NIST STANDARIZATION PROCESS

NIST (National Institute of Standards and Technology), founded in 1901, is one of the US's oldest physical science laboratories and has been a significant role in promoting equitable standards and creating critical measurement solutions. Due to the concern about quantum computing and its power to break many public-key cryptosystems, NIST started a formal process to develop post-quantum cryptography standards in 2016 [4]. It is soliciting proposals for public-key post-quantum cryptographic algorithms and up until July 22, 2020, the third round candidates were announced on its website. The seven third-round Finalists consist of two types of algorithms: public-key encryption and digital signatures. These candidates need to undergo a thorough evaluation and review before NIST makes a further decision. The new announcement is expected soon [3].

VI. CONCLUSION

It is uncertain when a practical quantum computer will be available, still the potential arrival of quantum computing technologies pushes for stronger cryptography, especially public-key encryption. To maintain information safety and security, we must begin now to prepare our information systems to have the ability to fight the quantum computing battle. International communities and organizations have started to take actions to address this issue. NIST has taken its responsibility for the development of guidelines and standards to improve and protect current information systems. Research teams from all around the world have participated in this task. The more we do today to be prepared, the easier our transition road will be when that day comes.

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